

Math 7243 Machine Learning - Homework 2

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Problem 1 Loss Functions:

a) Expanding the residual sum of squares,

$$\begin{aligned}\text{RSS}(\theta) &= (Y - X\theta)^T(Y - X\theta) \\ &= Y^TY - Y^TX\theta - \theta^TX^TY + \theta^TX^TX\theta.\end{aligned}$$

The term $Y^TX\theta$ is a scalar, so it equals its own transpose, θ^TX^TY . The two middle terms therefore combine:

$$\text{RSS}(\theta) = Y^TY - 2\theta^TX^TY + \theta^TX^TX\theta.$$

Differentiating with respect to θ ,

$$\frac{\partial}{\partial\theta}\text{RSS}(\theta) = -2X^TY + 2X^TX\theta.$$

Setting this equal to zero and dividing by 2,

$$X^TX\theta = X^TY.$$

Assuming X^TX is invertible, multiplying both sides on the left by $(X^TX)^{-1}$ gives the critical point

$$\theta = (X^TX)^{-1}X^TY$$

b) The Ridge loss is the RSS plus the penalty term,

$$\text{Ridge}_\lambda(\theta) = (Y - X\theta)^T(Y - X\theta) + \lambda\theta^T\theta.$$

The derivative of the first part is the one found in part a), and $\frac{\partial}{\partial\theta}\lambda\theta^T\theta = 2\lambda\theta$, so

$$\frac{\partial}{\partial\theta}\text{Ridge}_\lambda(\theta) = -2X^TY + 2X^TX\theta + 2\lambda\theta.$$

Setting this equal to zero and dividing by 2,

$$X^TX\theta + \lambda\theta = X^TY.$$

Writing $\lambda\theta = \lambda I\theta$ lets us factor out θ :

$$(X^TX + \lambda I)\theta = X^TY.$$

Since X^TX is positive semidefinite and $\lambda > 0$, the matrix $X^TX + \lambda I$ is positive definite and therefore always invertible. The critical point is

$$\theta = (X^TX + \lambda I)^{-1}X^TY$$

Problem 2 - Computing Linear Regression:

We fit $y = \theta_0 + \theta_1 x$, so the design matrix has a column of ones for the intercept:

$$X = \begin{bmatrix} 1 & 1.2 \\ 1 & 3.2 \\ 1 & 5.1 \\ 1 & 3.5 \\ 1 & 2.6 \end{bmatrix}, \quad Y = \begin{bmatrix} 7.8 \\ 1.2 \\ 6.4 \\ 2.6 \\ 8.1 \end{bmatrix}, \quad \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \end{bmatrix}.$$

Then

$$X^T X = \begin{bmatrix} 5 & 15.6 \\ 15.6 & 56.7 \end{bmatrix}, \quad X^T Y = \begin{bmatrix} 26.1 \\ 76.0 \end{bmatrix}.$$

a) Using the result of Problem 1.a), $\theta = (X^T X)^{-1} X^T Y$. The determinant of $X^T X$ is $5(56.7) - (15.6)^2 = 40.14$, so

$$\theta = \frac{1}{40.14} \begin{bmatrix} 56.7 & -15.6 \\ -15.6 & 5 \end{bmatrix} \begin{bmatrix} 26.1 \\ 76.0 \end{bmatrix} = \frac{1}{40.14} \begin{bmatrix} 294.27 \\ -27.16 \end{bmatrix} \approx \begin{bmatrix} 7.331 \\ -0.677 \end{bmatrix}.$$

The best fit function under the RSS loss is

$$\boxed{y \approx 7.331 - 0.677 x}$$

b) Using the result of Problem 1.b), $\theta = (X^T X + \lambda I)^{-1} X^T Y$.

For $\lambda = 1$:

$$X^T X + I = \begin{bmatrix} 6 & 15.6 \\ 15.6 & 57.7 \end{bmatrix}, \quad \det = 6(57.7) - (15.6)^2 = 102.84,$$

$$\theta = \frac{1}{102.84} \begin{bmatrix} 57.7 & -15.6 \\ -15.6 & 6 \end{bmatrix} \begin{bmatrix} 26.1 \\ 76.0 \end{bmatrix} = \frac{1}{102.84} \begin{bmatrix} 320.37 \\ 48.84 \end{bmatrix} \approx \begin{bmatrix} 3.115 \\ 0.475 \end{bmatrix}.$$

For $\lambda = 10$:

$$X^T X + 10I = \begin{bmatrix} 15 & 15.6 \\ 15.6 & 66.7 \end{bmatrix}, \quad \det = 15(66.7) - (15.6)^2 = 757.14,$$

$$\theta = \frac{1}{757.14} \begin{bmatrix} 66.7 & -15.6 \\ -15.6 & 15 \end{bmatrix} \begin{bmatrix} 26.1 \\ 76.0 \end{bmatrix} = \frac{1}{757.14} \begin{bmatrix} 555.27 \\ 732.84 \end{bmatrix} \approx \begin{bmatrix} 0.733 \\ 0.968 \end{bmatrix}.$$

The three fits are summarized below.

λ	Best fit function	$\theta^T \theta = \theta_0^2 + \theta_1^2$
0 (RSS)	$y \approx 7.331 - 0.677 x$	54.20
1	$y \approx 3.115 + 0.475 x$	9.93
10	$y \approx 0.733 + 0.968 x$	1.47

What explains the difference. The Ridge loss adds the penalty $\lambda \theta^T \theta = \lambda(\theta_0^2 + \theta_1^2)$, which charges for the size of θ . As λ increases, the minimizer gives up some fit to the data in exchange for a smaller θ , and the last column of the table shows $\theta^T \theta$ falling from 54.20 to 9.93 to 1.47.

Specifically, this penalty applies to the intercept θ_0 as well as the slope θ_1 . The RSS fit has a large intercept (7.331), which is the most heavily penalized part of θ , so Ridge pushes θ_0 toward zero. With the intercept forced down, the line can only stay near the data (whose averages are $\bar{x} = 3.12$ and $\bar{y} = 5.22$) by tilting upward, so the slope changes sign from negative to positive and grows. This is why θ_1 increases even though the penalty shrinks θ overall.

Problem 3

Stacking the data points, the model is $\vec{y} = X\vec{\theta}_* + \vec{\epsilon}$. We treat X as fixed, so the only randomness is the noise $\vec{\epsilon}$, whose entries are independent $\text{Normal}(0, \sigma^2)$. Hence $\mathbb{E}[\vec{\epsilon}] = \vec{0}$ and $\mathbb{E}[\vec{\epsilon}\vec{\epsilon}^T] = \sigma^2 I$.

Question 1: Substituting the model into the Ridge estimate,

$$\begin{aligned}\hat{\vec{\theta}} &= (X^T X + \lambda I)^{-1} X^T \vec{y} \\ &= (X^T X + \lambda I)^{-1} X^T (X\vec{\theta}_* + \vec{\epsilon}) \\ &= (X^T X + \lambda I)^{-1} X^T X \vec{\theta}_* + (X^T X + \lambda I)^{-1} X^T \vec{\epsilon}.\end{aligned}$$

The first term is not random. In the second term the matrix $(X^T X + \lambda I)^{-1} X^T$ is fixed, so by linearity of expectation

$$\mathbb{E}_D[(X^T X + \lambda I)^{-1} X^T \vec{\epsilon}] = (X^T X + \lambda I)^{-1} X^T \mathbb{E}[\vec{\epsilon}] = \vec{0}.$$

Therefore

$$\boxed{\mathbb{E}_D(\hat{\vec{\theta}}) = (X^T X + \lambda I)^{-1} X^T X \vec{\theta}_*}$$

Question 2: No. Writing $X^T X = (X^T X + \lambda I) - \lambda I$,

$$\begin{aligned}\mathbb{E}_D(\hat{\vec{\theta}}) &= (X^T X + \lambda I)^{-1} [(X^T X + \lambda I) - \lambda I] \vec{\theta}_* \\ &= \vec{\theta}_* - \lambda (X^T X + \lambda I)^{-1} \vec{\theta}_*.\end{aligned}$$

The bias is

$$\mathbb{E}_D(\hat{\vec{\theta}}) - \vec{\theta}_* = -\lambda (X^T X + \lambda I)^{-1} \vec{\theta}_*,$$

which is nonzero for $\lambda > 0$ (unless $\vec{\theta}_* = \vec{0}$). So $\hat{\vec{\theta}}$ is a *biased* estimator of $\vec{\theta}_*$: on average it is shrunk toward zero. Only when $\lambda = 0$ (ordinary least squares) is the bias zero.

Question 3: Let $A = (X^T X + \lambda I)^{-1} X^T$. From Question 1,

$$\hat{\vec{\theta}} - \mathbb{E}_D(\hat{\vec{\theta}}) = A\vec{\epsilon}.$$

Then

$$\begin{aligned}\text{Cov}(\hat{\vec{\theta}}) &= \mathbb{E}_D[(\hat{\vec{\theta}} - \mathbb{E}_D(\hat{\vec{\theta}}))(\hat{\vec{\theta}} - \mathbb{E}_D(\hat{\vec{\theta}}))^T] \\ &= \mathbb{E}[A\vec{\epsilon}\vec{\epsilon}^T A^T] \\ &= A \mathbb{E}[\vec{\epsilon}\vec{\epsilon}^T] A^T \\ &= \sigma^2 A A^T.\end{aligned}$$

Since $(X^T X + \lambda I)^{-1}$ is symmetric, $A^T = X(X^T X + \lambda I)^{-1}$, and so

$$\boxed{\text{Cov}(\hat{\vec{\theta}}) = \sigma^2 (X^T X + \lambda I)^{-1} X^T X (X^T X + \lambda I)^{-1}}$$

As a check, setting $\lambda = 0$ recovers the ordinary least squares covariance $\sigma^2 (X^T X)^{-1}$.